

Alex Jacobs

ajrjacob@uwaterloo.ca | +1 (437) 688-2039 | [LinkedIn](#) | [alexrijacobs.github.io](#)

Education

University of Waterloo

Sep. 2024 – Expected Jul. 2029

BASc Mechatronics Engineering — Robotics and Automation / cGPA 4.00/4.00

Waterloo, ON

Experience

Mechanical Designer — Quadrupedal & Humanoid Robotics

Sep. 2025 – Dec. 2025

Field AI

Irvine, CA

- Led end-to-end mechanical development of a modular, back-mounted sensing and compute payload for the Unitree G1 humanoid, porting existing hardware onto a new platform under tight timelines.
- Developed and validated ingress protection solutions, expanding the range of environments the robot could operate in.
- Designed and built modular test infrastructure, including a reconfigurable sensor tower rig, accelerating iteration on new sensor and payload configurations.
- Built sensor validation frameworks and apparatus to characterize components and guide part selection.

Research Assistant — Mobile Robotics & SLAM

Jan. 2025 – Apr. 2025

UCLA Sensing and Robotics for Infrastructure Lab

Los Angeles, CA

- Developed a custom Gazebo simulator in C++ and Python to collect LiDAR, visual, and inertial data from a quadrupedal robot performing waypoint navigation.
- Evaluated open-source SLAM implementations on datasets outside the domains they were designed for, characterizing where localization and mapping broke down.
- Debugged and tested proprietary laboratory software for robotic sensor data collection, processing, and visualization.
- Designed and 3D printed SolidWorks models to integrate new modules into ground and aquatic robot platforms.

Projects

Airborne Object Tracking System — Co-Founder & Mechanical Lead

Aug. 2025 – Present

Panopsys

Remote

- Co-founded a three-person venture building a ground-based system that localizes airborne objects in 3D by fusing detections from multiple distributed cameras, on a node-based ROS2 architecture.
- Own all mechanical design: camera and antenna mounting, sensor-head enclosure, and structural integration of compute and RF hardware on a portable tripod mast.
- Studied Kalman filtering to support the team's RTK-GNSS and IMU fusion, built on ROS2's UKF node, and reviewed modelling assumptions with the software lead.
- Led the system's first outdoor deployment at an airshow, capturing three hours of three-node data across 14 aircraft at ranges out to ~5 km.

Back-Mounted Sensing and Compute Payload

Nov. 2025 – Dec. 2025

Field AI

Irvine, CA

- Owned mechanical design, internal cable routing, and thermal integration of a rigid backpack payload for the Unitree G1 humanoid, from concept through fabricated prototype.
- Fabricated multiple payload iterations via 3D printing to enable parallel cross-office testing.
- Authored design documentation and file-structure guides that enabled clean handoff to downstream engineers.

Unmanned Surface Vehicle

Aug. 2023, Jan. 2025 – Apr. 2025

UCLA Sensing and Robotics for Infrastructure Lab

Los Angeles, CA

- Maintained and extended SolidWorks models of the platform's above- and below-water sensing kits.
- Modeled and 3D printed a hydrodynamic fairing to reduce drag around the vehicle's underwater sensors.
- Assisted with data collection using detachable above-water sensors for a visual-inertial navigation ablation study.

Technical Skills

CAD:	SolidWorks, Onshape, Fusion 360, Blender
Robotics & Software:	ROS2, Gazebo, C++, Python
Electronics & Embedded:	Microcontrollers (Teensy), sensor integration, wiring and harnessing
Manufacturing:	3D printing, lathe, milling machine, drill press, band saw, scroll saw

Airborne Object Tracking System

Panopsys / Co-Founder & Mechanical Lead / Aug. 2025 – Present



Node tracking a CP-140 Aurora during the airshow
test



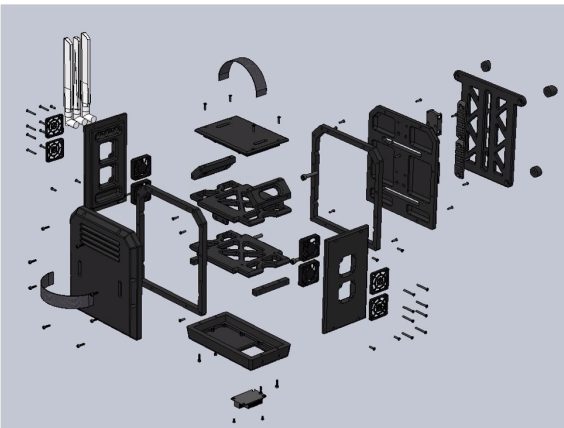
Camera module and radio hardware on the mast

First field deployment: 3 camera nodes | ~100m baselines | 14 display acts | overhead to ~5km | ~3h capture

- Co-founded a three-person venture developing a ground-based system that locates airborne objects in 3D by fusing detections from multiple distributed cameras, running on a node-based ROS2 architecture.
- Own all mechanical design: camera housings and mounts, antenna placement, sensor-head structure, and integration of compute, RF, and power hardware on a portable tripod mast.
- Designed the head for repeatable extrinsics and fast field setup, so cameras can be swapped or re-aimed between tests.
- Studied Kalman filtering to support the team's RTK-GNSS and IMU fusion, which runs on ROS2's UKF node, reviewing modelling decisions with the software lead.
- Ran the system's first outdoor deployment at an airshow: mechanical hardware held up all day, and the issues that surfaced were electrical and software-side, including an unseated antenna that blocked an RTK fix on a partner node. Dataset analysis is ongoing.
- Driving a v2 sensor head that replaces the 20mm extrusion frame with a single printed structure, removing the mounting constraints that dictated sensor placement on the prototype.

Back-Mounted Sensing and Compute Payload

Field AI / Mechanical Designer / Nov. 2025 – Dec. 2025



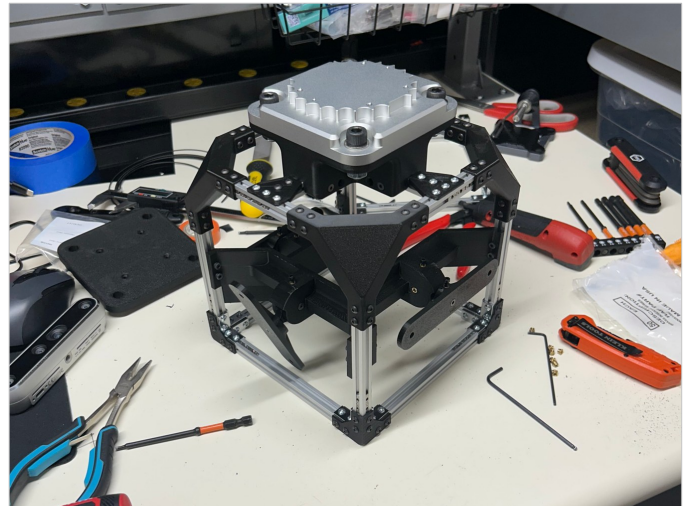
Exploded view of the payload assembly



Fabricated prototypes for cross-office testing

- Owned end-to-end mechanical design, wiring, and thermal integration of a rigid backpack payload for the Unitree G1 humanoid, letting existing payload hardware and software run on a new robotic platform.
- Architected internal cable routing and mounting so existing sensing and compute hardware dropped in cleanly, with serviceable access to connectors and fans.
- Designed and 3D printed multiple payload iterations for parallel cross-office testing, and wrote design and file-structure documentation for handoff to downstream engineers.

- Designed and manufactured a modular rigid sensor tower supporting LiDAR, camera, ultrasonic, and compute payload testing with configurable extrinsics.
- Built custom 3D printed fixtures and a Teensy 4.0-based validation framework to streamline sensor characterization and comparison.
- Applied rapid prototyping methods to deliver a robust yet reconfigurable platform that supports ongoing perception development rather than a single test.



Assembled sensor tower test rig

Unmanned Surface Vehicle

UCLA Sensing and Robotics for Infrastructure Lab / Aug. 2023, Jan. – Apr. 2025



The unmanned surface vehicle and its sensing kit

- Contributed to mechanical and software bring-up of an unmanned surface vehicle used for visual-inertial navigation research.
- Maintained and extended SolidWorks models of the above- and below-water sensing kits, including prototype hull components and camera mounts for evolving sensor configurations.
- Modeled and 3D printed a hydrodynamic fairing to reduce drag around the vehicle's underwater sensors.
- Assisted in debugging SLAM and data collection pipelines, improving the reliability of field experiments.

Simulation and SLAM Evaluation

UCLA Sensing and Robotics for Infrastructure Lab / Jan. – Apr. 2025

- Built a custom Gazebo simulator in C++ and Python to collect synchronized LiDAR, visual, and inertial data from a quadrupedal robot running waypoint navigation.
- Ran open-source SLAM implementations against datasets outside their intended domain and assessed where they held up, comparing reconstructed point clouds and localization behaviour to identify failure cases, and debugged the lab's data collection and visualization tooling.

Design & CAD:	SolidWorks, Onshape, Fusion 360, Blender — enclosures, mounts, tolerancing, exploded documentation
Fabrication:	FDM 3D printing, lathe, mill, drill press, band saw; design for rapid iteration
Robotics & Software:	ROS2, Gazebo, C++, Python, Kalman filtering fundamentals, SLAM evaluation
Test & Validation:	Teensy-based test frameworks, sensor characterization, field testing, ingress protection